CLIMATE ML · RAJASTHAN 2006–2025

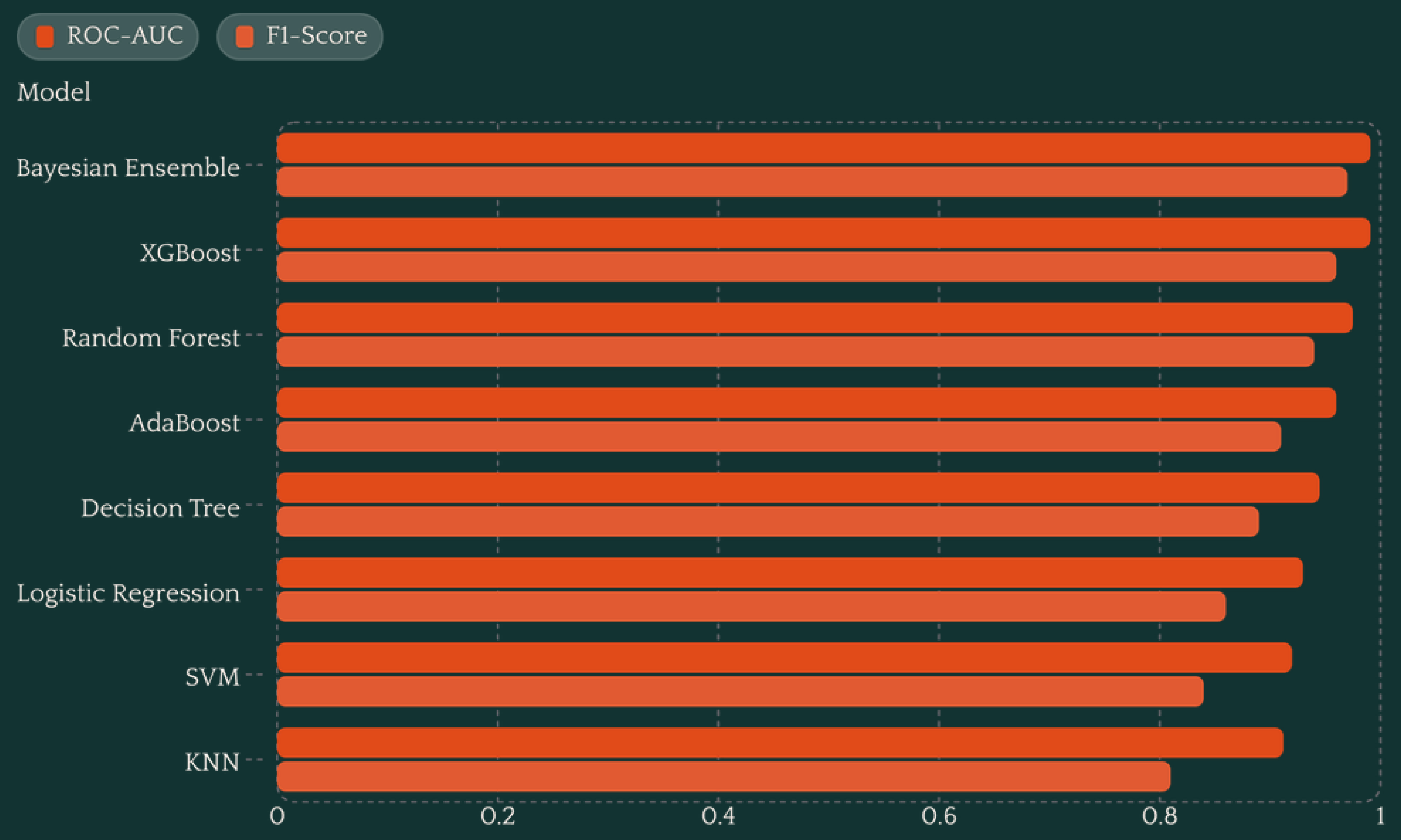
Binary Classification of Heatwave Events in Rajasthan

This project frames heatwave detection as a supervised binary classification problem — predicting whether a given day constitutes a heatwave event using historical meteorological data spanning 2006 to 2025. The pipeline enforces a leakage-free 70/15/15 train/val/test split, applies domain-specific feature engineering, and addresses class imbalance via SMOTE.

MODEL EVALUATION

Classification Performance Across 8 Models

Eight models were benchmarked across five standard metrics. Tree-based models and ensembles dominate, while distance-based and linear approaches show comparatively weaker discrimination on this high-dimensional meteorological feature space.



Best vs. Worst: What Drove the Gap?

🏆 WINNER

Bayesian Weighted Ensemble & XGBoost

ROC-AUC: 0.991

Gradient boosting naturally captures non-linear meteorological thresholds — temperature gradients, humidity interactions, and pressure anomalies — without manual tuning. The Bayesian ensemble further stabilizes predictions by weighting base learners by posterior confidence.

📉 LAGGARD

K-Nearest Neighbors

ROC-AUC: 0.912

KNN suffered from **high-dimensional distance degradation**: as feature dimensionality grows, Euclidean distances converge, eroding the model's ability to distinguish heatwave vs. non-heatwave neighborhoods. No inherent feature weighting compounds this weakness.

KEY TAKEAWAYS

From Model to Early Warning System

Ensembling Stabilizes Alerts

Bayesian-weighted ensembling of top tree classifiers significantly reduces variance in early heatwave warning outputs — critical for operational deployment where false negatives carry public health consequences.

Leakage Prevention Ensures Realistic Generalization

Strict temporal split discipline and feature-level leakage audits eliminated optimistic bias, ensuring test-set metrics reflect true out-of-sample performance on future unseen data.

Gradient Boosting Matches Meteorological Complexity

XGBoost's native handling of heterogeneous feature scales and threshold-based splits aligns naturally with how heatwave conditions emerge from interacting atmospheric variables.